

Scheduling Algorithm

1. fcfs

```
midhun@localhost: ~  
[midhun@localhost ~]$ ls  
bank Desktop davan Music Pictures sh Videos  
a banker.c Documents devan.c os_lab Public sjf while.sh  
abc bankers Downloads dev os_lab1 SEC sjf.c while.sh  
abc.sh bm h dev.c os_lab1.c sample.c Templates XYZ  
ab.sh close dev devan os_lab.c sample.txt test.sh  
[midhun@localhost ~]$ vi bestfit.c  
[midhun@localhost ~]$ gcc bestfit.c -o bestfit  
[midhun@localhost ~]$ ./bestfit  
Enter number of processes: 3  
Enter Arrival Time and Burst Time for P1: 0 5  
Enter Arrival Time and Burst Time for P2: 1 3  
Enter Arrival Time and Burst Time for P3: 2 8  
  
Best Fit Average Turnaround Time: 8.67  
Best Fit Average Waiting Time: 3.33  
[midhun@localhost ~]$
```

```
midhun@localhost: ~  
[midhun@localhost ~]$ ls  
.. bank Desktop matthu Music Pictures sh while.sh  
a devan.c Documents devan os_lab Public Templates XYZ  
sjf bankers Downloads mattere os_lab1 RFC trat.sh  
abc.sh hm h monb os_lab1.c sample.c RTOSes  
ab.sh close sjf mtch11 os_lab.c sample.tss while.sh  
[midhun@localhost ~]$ vi sjf  
[midhun@localhost ~]$ vi sjf.c  
[midhun@localhost ~]$ vi sjf.c  
[midhun@localhost ~]$ vi sjf.c  
[midhun@localhost ~]$ gcc sjf.c -o sjf  
[midhun@localhost ~]$ ./sjf  
Enter number of processes: 3  
Enter Arrival Time and Burst Time for P1: 8 7  
Enter Arrival Time and Burst Time for P2: 2 4  
Enter Arrival Time and Burst Time for P3: 4 1  
  
Average Turnaround Time: 7.00  
Average Waiting Time: 3.80  
[midhun@localhost ~]$
```

2.sjf

```
int midhun(char completiontimes && done[]) == 0)
{
    if(n[i] < a[i]);
    {
        minst = bt[i];
        idx = i;
    }
}

if(idx != -1)
{
    if(completiontime < at[idx])
        completiontime = at[idx];

    completiontime += bt[idx];

    tat[idx] = completiontime - at[idx];
    wt[idx] = tat[idx] - bt[idx];

    done[idx] = 1;
    completed++;
}
else
{
    completiontime++;
}
}
for(int i = 0; i < n; i++)
{
    avgtat += tat[i];
    avgwt += wt[i];
}
printf("\nAverage Turnaround Time: %.2f\n", avgtat / n);
printf("Average Waiting Time: %.2f\n", avgwt / n);

return 0;
}
```

```
#include <stdio.h>
#include <stdlib.h>
void main()
{
    int n;
    printf("Enter number of processes: ");
    scanf("%d", &n);

    int at[n], bt[n], tat[n], wt[n];
    int done[n];

    // Input
    for(int i = 0; i < n; i++)
    {
        printf("Enter Arrival Time and Burst Time for P%d: ", i+1);
        scanf("%d %d", &at[i], &bt[i]);
        done[i] = 0;
    }

    int completed = 0;
    int completionTime = 0;
    int min;
    int target = 0;

    // SJF Scheduling (Non-Preemptive)
    while(completed < n)
    {
        int idx = -1;
        int minb = 1e9;

        for(int i = 0; i < n; i++)
        {
            if(pr[i] == completionTime && done[i] == 0)
            {
                if(bt[i] < minb)
                {
                    minB_ = bt[i];
                    idx = i;
                }
            }
        }

        if(idx != -1)
```

```

int midhun(char completiontimes && done[]) == 0)
{
    if(n[i] < a[i]);
    {
        minst = bt[i];
        idx = i;
    }
}

if(idx != -1)
{
    if(completiontime < at[idx])
        completiontime = at[idx];

    completiontime += bt[idx];

    tat[idx] = completiontime - at[idx];
    wt[idx] = tat[idx] - bt[idx];

    done[idx] = 1;
    completed++;
}
else
{
    completiontime++;
}
}
for(int i = 0; i < n; i++)
{
    avgtat += tat[i];
    avgwt += wt[i];
}
printf("\nAverage Turnaround Time: %.2f\n", avgtat / n);
printf("Average Waiting Time: %.2f\n", avgwt / n);

return 0;
}

```

3.Round Robin

```

int midhun(char completiontimes && done[]) == 0)
{
    if(n[i] < a[i]);
    {
        minst = bt[i];
        idx = i;
    }
}

if(idx != -1)
{
    if(completiontime < at[idx])
        completiontime = at[idx];

    completiontime += bt[idx];

    tat[idx] = completiontime - at[idx];
    wt[idx] = tat[idx] - bt[idx];

    done[idx] = 1;
    completed++;
}
else
{
    completiontime++;
}
}
for(int i = 0; i < n; i++)
{
    avgtat += tat[i];
    avgwt += wt[i];
}
printf("\nAverage Turnaround Time: %.2f\n", avgtat / n);
printf("Average Waiting Time: %.2f\n", avgwt / n);

return 0;
}

```

```
Activities Terminal Mar 18 29:57
midhun@localhost:~

[midhun@localhost ~]$ ls
a      bank      best.c    mait      Music     Pictures  sh       Videos
b      banker.c  bestout   matrix    os_lab    Public    sjf      wrtle.sh
ABC    banker.o  bestout.txt matrix.o   os_lab1   REC       sjf.c    wrtio.sh
abc.sh bm        h         much      os_lab2.c saople.c  Templates xyz
ab.sh  close    mai       mtbtt     os_lab.c  sample.txt test.sh

[midhun@localhost ~]$ vi vi roundrobin.c
[midhun@localhost ~]$ ls gcc roundrobin.c -o roundrobin
[midhun@localhost ~]$ ls ./round robin
bash: ./round: No such file or directory
[midhun@localhost ~]$ -s ./roundrebin
Enter number of processes:
3
Enter Time Quantum:
2
Enter AT and BT for P1:
0 5
Enter AT and BT for P2:
1 3
Enter AT and BT for P3:
2 1

Average Turnaraund Time: 6.23
Average Waiting Time: 3.33
[midhun@localhost ~]$
```

```
Activities Terminal Mar 18 10:06
midhun@localhost:~/usr/bin/vim reundrebin.c

#include <stdio.h>
int main()
{
    int t, to, completed = 0, completionTime = 0;

    printf("Enter number of processes: \n");
    scanf("%d", &n);

    printf("Enter the Quantum: \n");
    scanf("%d", &tq);

    int at[n], bt[n], rem[n], tat[n], wt[n];
    float avgtat = 0, avgwt = 0;

    // Input
    for (int i = 0; i < n; i++)
    {
        printf("Enter AT and BT for P%d: \n", i + 1);
        printf("AT:"); scanf("%d", &at[i]);
        printf("BT:"); scanf("%d", &bt[i]); // Remaining time = Burst time
        rem[i] = bt[i];
    }

    // Round Robin Logic
    while (completed < n)
    {
        int i, done = 1;
        for (int i = 0; i < n; i++)
        {
            if (at[i] <= completionTime && rem[i] > 0)
            {
                done = 0;
                if (rem[i] > tq)
                    completionTime += tq;
                rem[i] -= tq;
            }
            else
            {
                completionTime += rem[i];
            }
        }
    }
}
```

1,1 Top

```
Activities Terminal Mar 18 19:05
midhun@localhost: ~/usr/bin/vim reundrobin.c

int ldte = -1;

for (int i = 0; i < n; i++)
{
    if (at[i] <= completionTime && done[i] == 0 && rem[i] > 0)
    {
        ldlo = i;
        if (rem[i] > tq)
        {
            completionTime += tq;
            rem[i] -= tq;
        }
        else
        {
            completionTime += rem[i];
            tat[i] = completionTime - at[i];
            wt[i] = tat[i] - bt[i];
            rem[i] = 0;
            done[i] = 1;
            completed++;
        }
    }
    if (ldte)
        completionTime++;
}

// Calculate averages
for (int i = 0; i < n; i++)
{
    avgtat += tat[i];
    argwt += wt[i];
}

printf("\nAverage Turnaround Time: %.2f\n", avgtat / n);
printf("Average Waiting Time: %.2f\n", argwt / n);

return 0;
```